



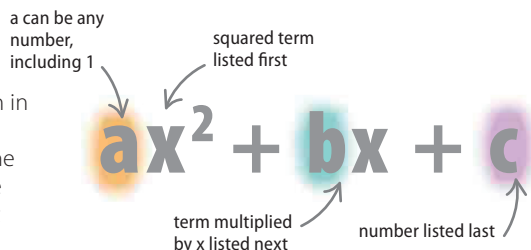
# Quadratic expressions

A QUADRATIC EXPRESSION CONTAINS AN UNKNOWN TERM (VARIABLE) SQUARED, SUCH AS  $x^2$ .

An expression is a collection of mathematical symbols, such as  $x$  and  $y$ , and operations, such as  $+$  and  $-$ . A quadratic expression typically contains a squared variable ( $x^2$ ), a number multiplied by the same variable ( $x$ ), and a number.

## What is a quadratic expression?

A quadratic expression is usually given in the form  $ax^2 + bx + c$ , where  $a$  is the multiple of the squared term  $x^2$ ,  $b$  is the multiple of  $x$ , and  $c$  is the number. The letters  $a$ ,  $b$ , and  $c$  all stand for different positive or negative numbers.



### SEE ALSO

◀ 174–175 Expanding and factorizing expressions

Factorizing quadratic equations 190–191 ▶

### ◀ Quadratic expression

The standard form of a quadratic expression is one with squared term ( $x^2$ ) listed first, terms multiplied by  $x$  listed second, and the number listed last.

## From two parentheses to a quadratic expression

Some quadratic expressions can be factorized to form two expressions within parentheses, each containing a variable ( $x$ ) and an unknown number. Conversely, multiplying out these expressions gives a quadratic expression.

### Multiplying two expressions in parentheses

means multiplying every term of one parenthesis with every term of the other. The final answer will be a quadratic equation.

To multiply the two parentheses, split one of the parentheses into its terms. Multiply all the terms of the second parenthesis first by the  $x$  term and then by the numerical term of the first parenthesis.

Multiplying both terms of the second parenthesis by each term of the first in turn results in a squared term, two terms multiplied by  $x$ , and two numerical terms multiplied together.

Simplify the expression by adding the  $x$  terms. This means adding the numbers together inside parentheses and multiplying the result by an  $x$  outside.

Looking back at the original quadratic expression, it is possible to see that the numerical terms are added to give  $b$ , and multiplied to give  $c$ .

